Nvidia Corp.

Equity Analysis

By Noah Rubin and Kyle Murphy

Recommendation

We have found that Nvidia Corp. is undervalued and recommend buying at its current share price of $142.44. We believe that the target price for Nvidia should be $224.37, which is a return of 57% of its current share price. We produced this price by using the Dividend Discount Model (DDM), the Relative Valuation Model, and analyst's target prices. We also believe that the stock should be held for 2 years.

Background Information

Overview

Nvidia is one of the largest and fastest growing technology companies in the world, which specializes in graphic processing units (GPUs) and artificial intelligence (AI). Nvidia operates two business segments: Compute & Networking and Graphics. Nvidia was founded in 1993 by Jensen Huang with the goal of advancing 3D graphics for personal computers. Since then, it has become one of the largest microchip processors in the world after its invention of the GPU in 1999. Its data center scale offerings have reshaped the technology industry, and it has used its GPU framework to build its own accelerated computing platforms, AI solutions, scientific computing, data science, robotics, metaverse and 3D internet applications. Although it has facilities throughout the world primarily in Taiwan, China, India and Israel, the US sector generates more than 45% of its revenue.

Operations

The Compute and Networking segment accounts for almost 80% of Nvidia's revenue and includes all their latest breakthroughs in the artificial intelligence sector. This sector comprises Nvidia's data center accelerated computing platform, automotive artificial intelligence, autonomous driving development, autonomous vehicle solutions, electric vehicle computing platforms, Jetson robotics, the DGX Cloud and other Nvidia AI enterprise software.

The graphics segment generates the other 20% of Nvidia's revenue which is surprising because this is what Nvidia is known for. The graphics segment includes the GeForce GPUs for gaming and PCs, GeForce NOW game streaming service and infrastructure, its solutions for various gaming platforms, virtual GPU software for cloud-based visual and virtual computing, autonomous infotainment systems and the Omniverse Enterprise software for building and operating 3D internet applications.

Industry Analysis

After the market for semiconductors took a massive hit due to shortages, geopolitical tensions and weak demand, there is optimism that a slow and steady recovery will continue to take place. Increasing valuations for AI companies signal optimism for a strong comeback and these companies are likely to continue displaying their strengths in chips, memory, and advanced-packaging providers. Some companies that Nvidia is competing against in the AI & Data Center sector are ARM Holdings US, Broadcom Inc, Marvell Technology Inc, Advanced Micro Devices (AMD) Inc and Intel Corporation.

Macroeconomic Effects

- **Economic Growth:** Nvidia has experienced significant growth that is heavily tied to broader economic trends, specifically in the technology and semiconductor sectors. The company being a top producer of graphics processing units and AI hardware has been rewarding as there has been increasing demand for advanced computing power across several industries including, but not limited to, artificial intelligence, data centers, gaming, and autonomous vehicles. The economic expansion in cloud computing, machine learning, and digital entertainment has stimulated the demand for Nvidia to continue producing new innovative products. As governments and businesses push towards more developed economies driven by artificial intelligence, Nvidia will only see more growth if they continue to meet the rising demand of the economy.
- **Interest Rates:** Changing interest rates can have crucial effects on the borrowing power of not only consumers but also for businesses looking to raise capital. With the Fed slashing interest rates lately, Nvidia can borrow more money to help its vast arsenal of modern technology.
- **Technology Trends:** Nvidia is an industry leader in the innovative technology industry, but if another company makes a breakthrough, Nvidia will look to develop it even further. The markets are looking to expand AI usage for everyday life, and this is a key strength that Nvidia has.

Financial Performance

- NVDA reported a 125.9% growth in revenue at $60.922 billion for the 2024 fiscal year.
- In 2024, the gross profit margin of NVDA jumped 14.9% to 73.6% from the previous year while their operating margin also increased 30% to 55.1%.
- Nvidia's adjusted EPS improved 10.03% year over year to $6.74.
- The company reported $25.98 billion in cash at the beginning of the 2024 fiscal year, up 95.42% from 2023.

Investment Thesis

Key Drivers: Nvidia is a dominant performer in several industries including semiconductors and artificial intelligence. Their performance is largely driven by their ability to harness the opportunities of artificial intelligence as it has led them towards data center accelerated computing platforms, autonomous driving technologies, and enterprise AI software. This strong foothold in artificial intelligence has helped them drive towards significant growth in a rapidly expanding AI market.

Competitive Advantage: Nvidia's competitive advantage lies heavily in its proprietary technology, well defined intellectual property portfolio, and brand awareness. The company has a strong reputation for providing quality high-performance GPUs as well as the ability to adapt quickly in the expanding AI market. Nvidia's GPU architecture, powering both consumer-facing products like GeForce gaming GPUs and enterprise-level solutions such as the DGX Cloud and autonomous driving platforms separates them from their competitors. The company also holds thousands of patents on their GPU technology, protecting its innovations. Lastly, Nvidia has been able to partner with some of the leading tech companies and establish their name as a brand, creating even more of a gap between itself and its competitors.

Sustainable Growth: The company's ability to maintain strong revenue growth and profitability measures are a positive sign of sustainable growth. Nvidia's strong margins combined with its competitive positioning within the market show the potential for its long-term growth. Beyond their financial performance they hold key intangible assets and have the ability to quickly innovate in a versatile market.

Investment Catalysts

Short-Term Catalysts

1. **Quarterly Earnings Reports and Strong Financial Performance:** Nvidia's quarterly earnings, especially their growth in revenues from gaming and data center programs, are key short-term catalysts. Their ability to continuously show strong earnings and financial performance remains a critical driver of short-term stock price movement.
2. **New Product Launches and Innovations:** Nvidia has regularly released new product lines, particularly GPUs and AI hardware solutions. The launch of new graphics cards, or updates in their data center products, are pivotal for driving immediate revenue and market share expansion. They have also been able to meet the increasing demand of the artificial intelligence market through their innovation.

On October 12th, 2022, Nvidia released its GeForce RTX 40 SUPER Series, a much more advanced GPU than its predecessor. This release boosted Nvidia's short term performance by increasing the demand for high quality GPUs, generating additional revenue, and increasing investor confidence.

Long-Term Catalysts

3. **Dominance in AI and Deep Learning Technology:** The rapid growth of artificial intelligence and deep learning is a key long-term catalyst for NVIDIA. As industries such as healthcare, financial services and automotive continue to look for new ways to evolve, Nvidia will play a key role in furthering the AI adoption in these industries. NVIDIA's ongoing leadership in AI, especially its CUDA software, and partnerships with leading tech companies, has helped prove how valuable the company is.
4. **Nvidia's Entrance into the Auto Industry:** NVIDIA's automotive business, NVIDIA DRIVE, is an important long-term catalyst. As the global automotive industry moves toward all electric vehicles and autonomous driving, NVIDIA stands to benefit from its partnerships with major automakers. The integration of NVIDIA's AI technologies in next-generation vehicles will help create sustained revenue streams and growth opportunities.

On December 3rd, 2024, Nvidia released its breakthroughs in creating robot brains to allow robots to autonomously handle a wide range of industrial processes and tasks. "Isaac Sim is a reference application built on NVIDIA Omniverse for developers to simulate and test AI-driven robots in physically based virtual environments."

Valuation

To determine the target price for NVDA, we utilized two valuation methods: (1) the Dividend Discount Model (DDM) and (2) the Relative Valuation Model. Additionally, we factored in analyst recommendations to provide a well-rounded assessment.

DDM Model

Exhibit A:

NVIDIA Corp

Model Assumptions			
		Risk Premium Country/Region	United States
Earnings Per Share FY1	2.927	Bond Rate	4.199 %
Earnings Per Share FY2	4.349	Country/Region Premium	5.578 %
Earnings Per Share FY3	5.327	Beta	1.854
Dividends Per Share FY1	0.039	1) Risk Premium	10.341 %
Growth Years	5	Payout during Growth yrs	1.332 %
Transitional Years	12	Payout at Maturity	45.000 %
Long Term Growth Rate	49.810 %	Growth Rate at Maturity	7.997 %
Closing Price	138.805	Currency	USD

Computed Values	
Theoretical Price	405.387
Percentage Change from Cl...	192.055 %
Internal Rate of Return	19.157 %
Expected Return	91.867 %
Implied Growth Rate	35.028 %

Relative Valuation Model

Exhibit B:

Analysis of NVDA US Multiples - Premium to Comps

3M 6M 1Y 2Y 5Y

Metric	Current vs 2Y Average Historical Premium: Current	Hist Avg	Diff	# SD	3M Trend	2Y Historical Premium Range: Low	Range	High	Implied @ Hist Avg: Multiple	Price (USD)
Current Price							● Current ◆ Hist Avg			138.50
1) BF P/E	8%	40%	-32%	-0.9		-8%		113%	43.6x	179.23
2) BF EV/EBITDA	34%	88%	-54%	-0.8		9%		225%	38.4x	193.93
3) BF EV/EBIT	1%	35%	-34%	-0.9		-12%		112%	37.8x	184.11
4) BF EV/Rev	--	--	--	--		82%		204%	--	--
5) LF P/BV	--	--	--	--		--		--	--	--

Analyst Recommendations

Exhibit C:

ANR NVIDIA CORP Equity | Options

NVIDIA CORP Equity | ANR | Related Functions Menu | Message

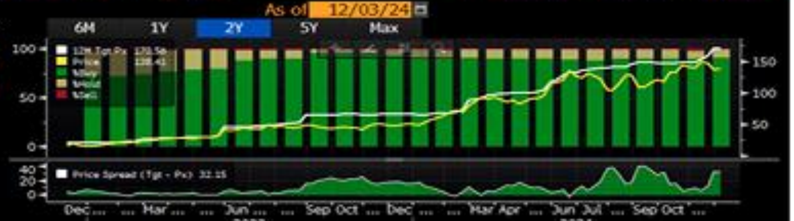
NVDA US $ ↑ 138.42 -0.21 S138.40 / 138.42S 7x7

At 10:41 d Vol 48,768,041 O 138.26Z H 140.54D L 137.95Z Val 6.786B

NVDA US Equity | Actions | Alert | Export | Analyst Recommendations

NVIDIA Corp — As of 12/03/24

Consensus Rating	▲	4.78
Buys	90.8%	69
Holds	9.2%	7
Sells	0.0%	0
12M Tgt Px	56/82	170.56
Last Price		138.41
Pricing Currency		USD
Return Potential		23.2%
LTM Return		196.0%



Entitled to 82 of 87 sources

	Firm	Analyst	Recommendation	Tgt Px	Date	1 Yr Rtn	BARR	Rank
1)	Mizuho Securities	Vijay R Rakesh	outperform	175.00	12/02/24	196.53%	2	
2)	Evercore ISI	Mark Lipacis	outperform	190.00	12/02/24			
3)	Melius Research LLC	Benjamin A Reitzes...	buy	195.00	12/02/24	196.53%	2	
4)	SinoPac Securities	Team Coverage	buy	170.00	12/01/24			
5)	BNP Paribas Exane	David O'Connor	outperform	170.00	11/29/24	196.53%	2	10
6)	Bernstein	Stacy Aaron Rasgon	outperform	175.00	11/28/24	196.53%	2	
7)	First Shanghai Sec...	Peter Han	buy	160.00	11/28/24	196.53%	2	
8)	Baptista Research	Ishan Majumdar	↑ buy	173.30	11/28/24	87.02%		
9)	Jefferies	Blayne Curtis	buy	185.00	11/26/24			
10)	President Capital M...	Darren Chen	buy	175.00	11/26/24	68.52%		
11)	Punto Research	Adair Casco	↑ hold	140.00	11/26/24	47.28%		
12)	KeyBanc Capital Ma...	John Vinh	overweight	180.00	11/25/24	196.53%	2	8

Suggested Functions | GP Chart securities & technical studies | GPTR Chart & analyze insider transactions

	Weight	Target Price
DDM	20%	$405.39
Relative Valuation Model	45%	$185.76
Analyst Predictions	35%	$170.56
Target Price		**$224.37**

Intangible Assets

Intellectual Property: Nvidia owns thousands of patents on its products as they produce quite a wide range of products globally and have been developing their technology since 1993. Their graphics cards have been some of the best in the world ever since they invented them and these speculative technologies take a lot of patents in order to make sure no one steals the card's design. Nvidia also has a number of trade secrets, proprietary technologies and algorithms that are not available to the public yet as they are still in testing.

Brand Awareness: Nvidia has extremely strong brand awareness as one of the top technology companies in the world and this is shown by its stock performance in the last few years. The company has relationships with not only ordinary customers buying gaming computers, but also with large hardware manufacturers and governments. Nvidia has also a lot of money invested in research and development of new technology which is something that the company is known for.

Environmental Social Governance (ESG)

Exhibit D:

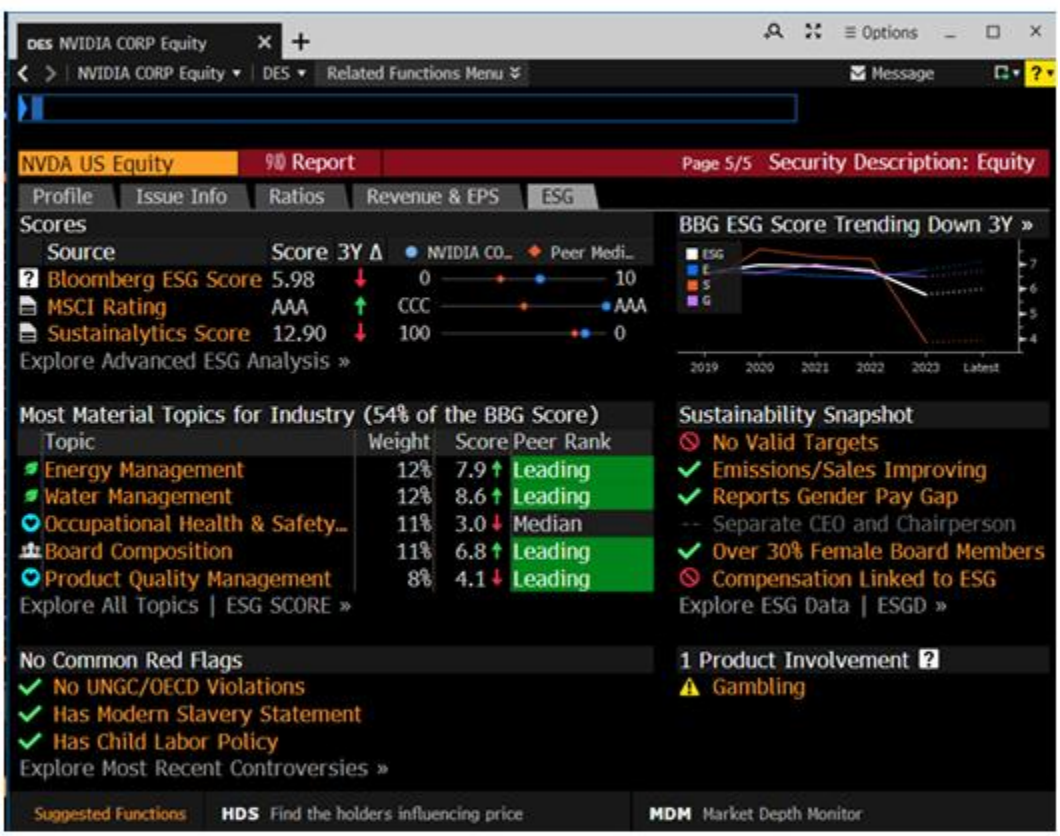
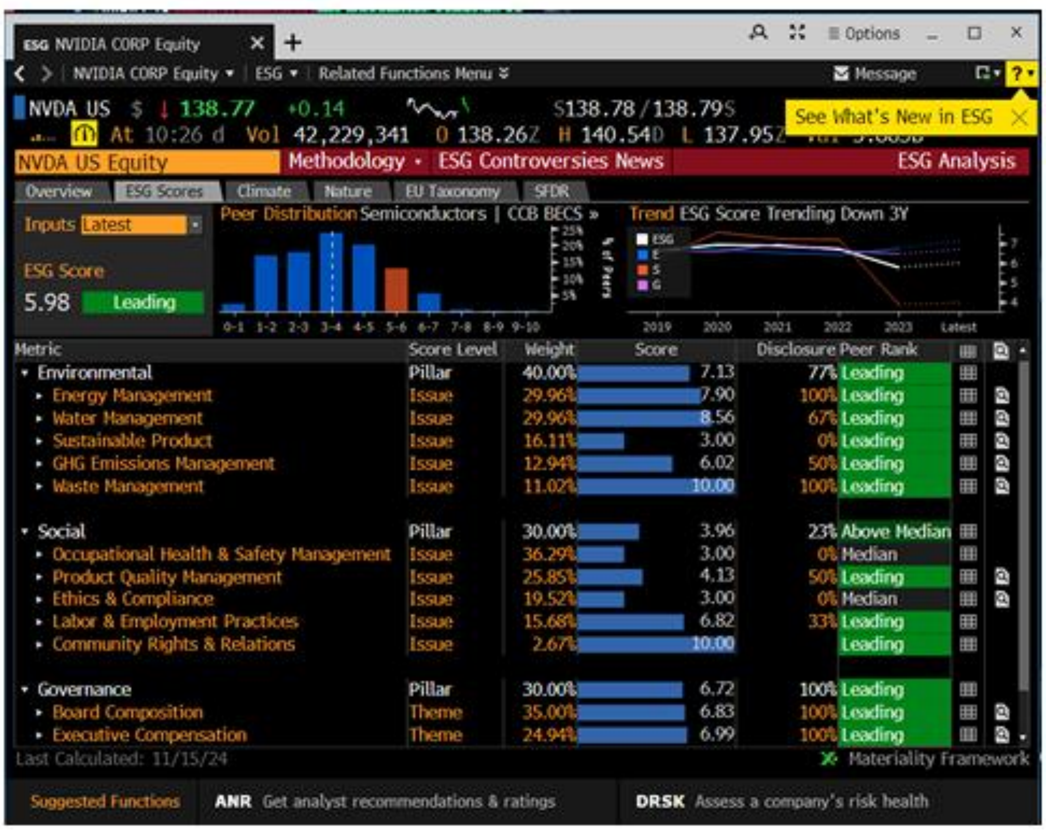


Exhibit E:

Exhibit F:



Risk Factors (10-K)

Global Markets: Changes in the demand for technology can impact Nvidia heavily, but since the world is looking to digitize society and Nvidia has a large global presence, this is something the company should not worry too much about. Something that did hit Nvidia hard was global supply chain issues. Nvidia has a large dependence on a global supply chain especially for semiconductors, which exposes its risk to economic downturns like during the Covid-19 pandemic. This is a large reason its Beta is 1.86, which is very high compared to the market. This means that the investment in Nvidia is quite risky, but it is worthy to note that Nvidia has seen an increase in its share price of over 50% in the past year.

Regulation and Legal Risk: Nvidia has been looked into several times for Antitrust violations in the UK and in the US. Since it is said that Nvidia has a monopoly on AI chips, this might lead to government fines and regulations that hurt the business.

Competition: Nvidia faces fierce competition in companies like Intel and AMD who are playing catch up with Nvidia in terms of variety of products and in technological advancements.

Mitigation

Nvidia has done so well recently because it has invested in a wide range of products and sectors of the technology industry. It is an industry leader and now knows how to stay ahead of the competition.